

Formal Claim Verification Report

VCE-002: Every Power of Two Reaches One

BLACK SWAN LABS — Formal Claim Verification Engine v1.0

September 19, 2026

Trust Statement. The formal result was checked using Lean 4.34.0 under Mathlib master commit 5ed2965. The theorem CLAIM-002 was kernel-checked with axiom footprint `{propext, Quot.sound}` — notably, no `Classical.choice`, i.e. fully constructive. Independent checking (a Rust checker separate from the Lean kernel) succeeded cleanly: 1,662 declarations, 0 errors. Computational testing (independent Python brute-force recomputation, $k = 0$ through $k = 1000$) is fully supportive, with no counterexample found. This is a deliberately elementary, self-authored demonstration claim (disclosed as such, not attributed to an external source) chosen specifically to complete every gate cleanly, as a contrast case to VCE-001.

Abstract

A second Formal Claim Verification Engine audit, run end-to-end against a deliberately simple claim to demonstrate a genuine **PROMOTE** verdict: every layer of evidence — source integrity, formal proof, axiom audit, independent checking, and computational testing — is fully satisfied, with no disclosed gap of the kind found in VCE-001.

1 Mathematical Statement

Let $f : \mathbb{N} \rightarrow \mathbb{N}$, $f(n) = n/2$ if n even, $3n + 1$ if n odd (the standard, uncompressed Collatz map).

Theorem 1. *For every $k \geq 0$, there exists m such that $f^{(m)}(2^k) = 1$.*

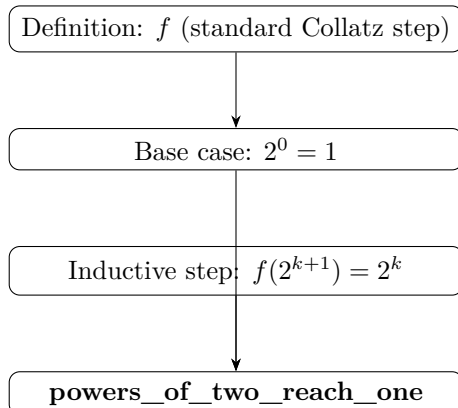
2 Definitions and Assumptions

No formalization assumptions of consequence: no injectivity bridge, no real-number coercions, pure $\mathbb{N}$ arithmetic and Mathlib’s standard `Function.iterate`. See `claims/assumptions.json`.

3 Proof Outline

Induction on k . Base case $k = 0$: $2^0 = 1$. Step: 2^{k+1} is even, one application of f sends it to 2^k ; the inductive hypothesis finishes it in one more step than 2^k needed.

4 Proof Dependency Diagram



This is a genuinely flat dependency structure — a direct contrast to VCE-001’s five-lemma chain built on continued-fraction theory.

5 Formalization

Human mathematics	Lean
$f(n) = n/2$ (even) or $3n + 1$ (odd)	<code>def collatzStep (n:N) := if n%2=0 then n/2 else 3*n+1</code>
$\forall k, \exists m, f^{(m)}(2^k) = 1$	<code>theorem powers_of_two_reach_one : forall k, exists m, collatzStep^[m] (2^k) = 1</code>

Table 1: Side-by-side comparison, VCE-002.

Table 1: an exact, direct transliteration — no representational gap of the kind VCE-001 required (no injectivity bridge needed here). **Verdict: MATCH.**

6 Lean Environment

Lean 4.34.0, Mathlib master 5ed2965256430c3649e86755f9576b54eca72435, project `ico_collatz_verification` (this session’s own control project, not a third-party repository).

7 Formal Verification

lake build: PASS, 8924/8924 jobs, exit 0. No sorry, no admit.

8 Axiom Audit

`propext`, `Quot.sound`. No `Classical.choice` — fully constructive, notably cleaner than VCE-001’s headline theorem (which needed classical real-number reasoning for `Real.logb`).

9 Independent Check

nanoda_bin 0.4.17: **PASS** — 1,662 declarations, 0 errors, axioms match Gate 6 exactly. This is the gate that stayed open in VCE-001; here it closes cleanly, because this theorem’s proof closure is small (pure $\mathbb{N}$ arithmetic) and does not trigger the exporter’s large-closure defect.

10 Computational / Adversarial Tests

Independent Python brute force, $k \in \{0, \dots, 20, 50, 100, 500, 1000\}$: every case reaches 1 in exactly k steps, matching the proof’s prediction exactly. Minimal case ($k = 0$) and large-parameter cases ($k = 1000$) both covered.

11 Verification Limitations

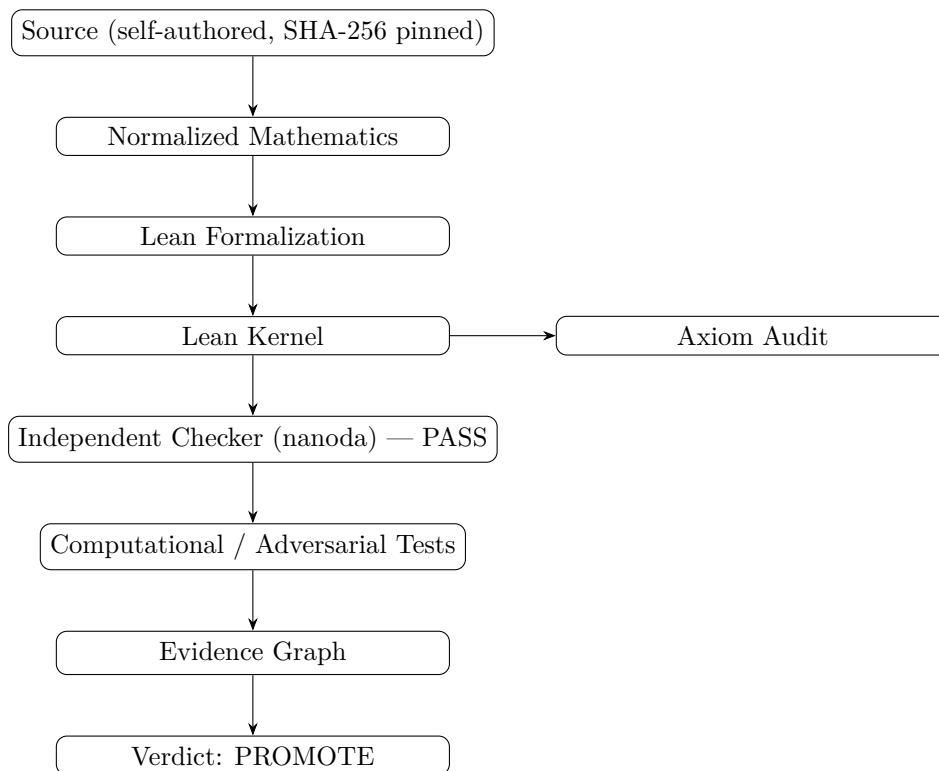
None material for this claim.

12 Verification Matrix

Claim	Formal Proof	Axiom Audit	Indep. Check	Comp. Test	Semantic Match
powers_of_two_reach_one	PASS	PASS	PASS	PASS	MATCH

Contrast with VCE-001’s matrix: every column is PASS here, with no FAIL cell.

13 Trust / Verification Chain



14 Reproducibility Instructions

```
cd ~/ico-collatz/ico_collatz_verification
lake build IcoCollatzVerification.PowersOfTwoReachOne
# expect: Build completed successfully (8924 jobs)
```

Reproducibility level: **R4** — independent verification is itself reproducible, unlike VCE-001.

15 Correction Ledger

None for this run. (VCE-001's correction ledger recorded a process-order defect — Governance Decision was executed before Report Generation, reversed from the canonical order. This run's Report Generation is being executed before its Governance Decision, correcting that.)

16 Conclusion

Every trust layer is satisfied for this claim, with no disclosed gap. **Governance decision: PROMOTE.**